

Abstract Algebra: Lecture Notes

Group Actions, the Class Equation, and Fundamentals of Ring Theory

Comprehensive Reference Covering Derived Subgroups, Actions, Conjugacy, Rings, Domains, and Fields

1. Preliminaries: Derived Subgroups of S_4 and A_4

Proposition 1.1. *The derived subgroup (commutator subgroup) of the symmetric group S_4 is the alternating group A_4 , that is, $S_4' = A_4$.*

Proof. Recall that for any two permutations $\sigma, \tau \in S_4$, the commutator is defined by $[\sigma, \tau] = \sigma\tau\sigma^{-1}\tau^{-1}$. Every cycle of length 3 can be expressed as a commutator of transpositions. Specifically, for distinct elements $a, b, c \in \{1, 2, 3, 4\}$, consider the transpositions (ab) and (bc) :

$$[(ab), (bc)] = (ab)(bc)(ab)^{-1}(bc)^{-1} = (ab)(bc)(ab)(bc) = (acb) = (abc)^2.$$

Because $(abc) = (acb)^2$, every 3-cycle is a commutator in S_4 . Since the set of all 3-cycles generates the alternating group A_4 , it follows that:

$$A_4 \subseteq S_4'.$$

Conversely, since the quotient group $S_4/A_4 \cong \mathbb{Z}_2$ is abelian, the universal property of commutator subgroups implies that $S_4' \subseteq A_4$. Combining both inclusions gives:

$$S_4' = A_4. \quad \square$$

Proposition 1.2. *The derived subgroup of the alternating group A_4 is the Klein four-group V_4 , that is, $A_4' = V_4$.*

Proof. Recall that the normal subgroups of A_4 are precisely:

$$\{I\}, \quad V_4, \quad \text{and} \quad A_4,$$

where $V_4 = \{I, (12)(34), (13)(24), (14)(23)\}$.

Since A_4 is non-abelian, its derived subgroup must be non-trivial:

$$A_4' \neq \{I\}.$$

Now consider the quotient group A_4/V_4 . The order of this quotient is:

$$|A_4/V_4| = \frac{|A_4|}{|V_4|} = \frac{12}{4} = 3.$$

Since any group of prime order is cyclic, A_4/V_4 is cyclic and hence abelian. By the characterization of derived subgroups, if the quotient A_4/N is abelian, then $A_4' \subseteq N$. Setting $N = V_4$, we obtain:

$$A_4' \subseteq V_4.$$

Because A_4' is a normal subgroup of A_4 contained in V_4 , and $A_4' \neq \{I\}$, the only possibility among the normal subgroups of A_4 is:

$$A_4' = V_4. \quad \square$$

2. Group Actions and Permutation Representations

Definition 2.1 (Left Group Action). Let G be a group and let A be a non-empty set. A **left group action** of G on A is a function

$$* : G \times A \longrightarrow A, \quad (g, a) \longmapsto g * a$$

satisfying the following two axioms:

- 1) **Associativity (Compatibility):** $g_1 * (g_2 * a) = (g_1 g_2) * a$ for all $g_1, g_2 \in G$ and all $a \in A$.
- 2) **Identity:** $e * a = a$ for all $a \in A$, where e is the identity element of G .

When such an action is given, A is called a G -set, and we say that G acts on A from the left.

Definition 2.2 (Right Group Action). A **right group action** of G on A is a mapping $* : A \times G \rightarrow A$ satisfying:

- 1) $(a * g_1) * g_2 = a * (g_1 g_2)$ for all $g_1, g_2 \in G$ and all $a \in A$.
- 2) $a * e = a$ for all $a \in A$.

Standard Examples of Group Actions

Example 2.3 (Trivial Action). Let G be any group and $A \neq \emptyset$. Define $* : G \times A \rightarrow A$ by:

$$g * a = a \quad \forall g \in G, \forall a \in A.$$

Then for any $g_1, g_2 \in G$ and $a \in A$, we have $g_1 * (g_2 * a) = g_1 * a = a$ and $(g_1 g_2) * a = a$, so associativity holds. Moreover, $e * a = a$. Hence, $*$ is a valid group action, termed the **trivial action**.

Example 2.4 (Action of a Group on Itself by Left Translation). Let $A = G$ and define $* : G \times A \rightarrow A$ by left group multiplication:

$$g * a = ga \quad \forall g \in G, \forall a \in A (= G).$$

Verifying the axioms:

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (g_2 a) = g_1 (g_2 a) = (g_1 g_2) a = (g_1 g_2) * a, \\ e * a &= ea = a. \end{aligned}$$

Therefore, $*$ is a group action, known as the **left regular action** or **left translation**.

Example 2.5 (Right Multiplication and Inversion). If one attempts to define $*$: $G \times A \rightarrow A$ on $A = G$ by $g * a = ag$, then:

$$g_1 * (g_2 * a) = g_1 * (ag_2) = (ag_2)g_1 = a(g_2g_1).$$

However, $(g_1g_2) * a = a(g_1g_2)$. Unless G is abelian, $a(g_2g_1) \neq a(g_1g_2)$, so this does **not** define a left action.

To define a left action using right multiplication, define $*$: $G \times A \rightarrow A$ by:

$$g * a = ag^{-1} \quad \forall g \in G, \forall a \in A (= G).$$

Verifying the axioms:

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (ag_2^{-1}) = (ag_2^{-1})g_1^{-1} = a(g_2^{-1}g_1^{-1}) = a(g_1g_2)^{-1} = (g_1g_2) * a, \\ e * a &= ae^{-1} = ae = a. \end{aligned}$$

Thus, this is a valid left action, called the **right regular action** (via inversion).

Example 2.6 (Action by Conjugation). Let $A = G$ and define $*$: $G \times A \rightarrow A$ by conjugation:

$$g * a = gag^{-1} \quad \forall g \in G, \forall a \in A (= G).$$

Verifying the axioms:

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (g_2ag_2^{-1}) = g_1(g_2ag_2^{-1})g_1^{-1} = (g_1g_2)a(g_1g_2)^{-1} = (g_1g_2) * a, \\ e * a &= eae^{-1} = a. \end{aligned}$$

Hence, $*$ is a group action, known as the **action by conjugation**.

Example 2.7 (Action on Left Cosets). Let $H \leq G$ and let $A = G/H = \{xH \mid x \in G\}$ be the collection of all left cosets of H in G . Define $*$: $G \times A \rightarrow A$ by:

$$g * (xH) = (gx)H \quad \forall g \in G, \forall xH \in A.$$

Verifying the axioms:

$$\begin{aligned} g_1 * (g_2 * xH) &= g_1 * ((g_2x)H) = (g_1(g_2x))H = ((g_1g_2)x)H = (g_1g_2) * xH, \\ e * (xH) &= (ex)H = xH. \end{aligned}$$

Thus, left multiplication on cosets constitutes a valid group action.

Example 2.8 (Conjugation on Subgroups). Let $A = \{H \mid H \leq G\}$ be the set of all subgroups of G . Define $*$: $G \times A \rightarrow A$ by:

$$g * H = gHg^{-1} \quad \forall g \in G, \forall H \in A.$$

Since gHg^{-1} is always a subgroup of G , $*$ maps into A . Verifying the axioms:

$$\begin{aligned} g_1 * (g_2 * H) &= g_1 * (g_2Hg_2^{-1}) = g_1(g_2Hg_2^{-1})g_1^{-1} = (g_1g_2)H(g_1g_2)^{-1} = (g_1g_2) * H, \\ e * H &= eHe^{-1} = H. \end{aligned}$$

Thus, $*$ is a valid group action of G on the set of its subgroups.

Remark 2.9 (Well-Definedness Property). If $*$: $G \times A \rightarrow A$ is a group action, then because $*$ is a function, whenever $x, y \in A$ satisfy $x = y$, it follows that $(g, x) = (g, y)$, and therefore:

$$g * x = g * y \quad \forall g \in G.$$

Equivalence with Permutation Representations

Theorem 2.10 (Action–Homomorphism Correspondence). *Let G be a group and A a non-empty set. Every homomorphism $f : G \rightarrow \text{Sym}(A)$ defines a group action of G on A . Conversely, every group action of G on A induces a unique homomorphism $f : G \rightarrow \text{Sym}(A)$.*

Proof. ($\implies$) Let $f : G \rightarrow \text{Sym}(A)$ be a homomorphism. For each $g \in G$, let $\sigma_g = f(g)$, where $\sigma_g : A \rightarrow A$ is a bijection. Since f is a homomorphism, for all $g_1, g_2 \in G$:

$$f(g_1 g_2) = f(g_1) \circ f(g_2) \implies \sigma_{g_1 g_2} = \sigma_{g_1} \circ \sigma_{g_2}.$$

Define $*$: $G \times A \rightarrow A$ by $g * a = \sigma_g(a)$ for all $a \in A$. Then:

$$g_1 * (g_2 * a) = g_1 * (\sigma_{g_2}(a)) = \sigma_{g_1}(\sigma_{g_2}(a)) = (\sigma_{g_1} \circ \sigma_{g_2})(a) = \sigma_{g_1 g_2}(a) = (g_1 g_2) * a.$$

Furthermore, since f is a homomorphism, $f(e) = I_A$ (the identity map on A), so:

$$e * a = \sigma_e(a) = I_A(a) = a.$$

Hence, $*$ is a group action of G on A .

($\impliedby$) Conversely, suppose $*$: $G \times A \rightarrow A$ is a group action. For each $g \in G$, define $\sigma_g : A \rightarrow A$ by:

$$\sigma_g(a) = g * a \quad \forall a \in A.$$

We first prove that σ_g is a bijection, so that $\sigma_g \in \text{Sym}(A)$:

- **Injectivity:** Let $a_1, a_2 \in A$ such that $\sigma_g(a_1) = \sigma_g(a_2)$. Then $g * a_1 = g * a_2$. Acting on the left by g^{-1} :

$$g^{-1} * (g * a_1) = g^{-1} * (g * a_2) \implies (g^{-1}g) * a_1 = (g^{-1}g) * a_2 \implies e * a_1 = e * a_2 \implies a_1 = a_2.$$

Thus, σ_g is one-to-one.

- **Surjectivity:** For any $a \in A$, observe that $g^{-1} * a \in A$, and:

$$\sigma_g(g^{-1} * a) = g * (g^{-1} * a) = (gg^{-1}) * a = e * a = a.$$

Thus, σ_g is onto.

Therefore, $\sigma_g \in \text{Sym}(A)$. Now define $f : G \rightarrow \text{Sym}(A)$ by $f(g) = \sigma_g$. For any $g_1, g_2 \in G$ and $a \in A$:

$$\sigma_{g_1 g_2}(a) = (g_1 g_2) * a = g_1 * (g_2 * a) = \sigma_{g_1}(\sigma_{g_2}(a)) = (\sigma_{g_1} \circ \sigma_{g_2})(a).$$

Consequently, $\sigma_{g_1 g_2} = \sigma_{g_1} \circ \sigma_{g_2}$, which means $f(g_1 g_2) = f(g_1) \circ f(g_2)$. Hence f is a homomorphism. $\square$

Definition 2.11 (Permutation Representation). The homomorphism $f : G \rightarrow \text{Sym}(A)$ induced by a group action is called the **permutation representation** associated with the action.

Corollary 2.12 (Cayley's Theorem). *Every group G is isomorphic to a subgroup of its symmetric group $\text{Sym}(G)$.*

Proof. Consider the left regular action of G on itself ($A = G$) defined by $g * a = ga$. By Theorem 2.10, this action induces a permutation representation:

$$f : G \longrightarrow \text{Sym}(G), \quad f(g) = \sigma_g, \quad \text{where } \sigma_g(a) = ga.$$

We examine the kernel of f . Suppose $f(g_1) = f(g_2)$. Then $\sigma_{g_1} = \sigma_{g_2}$, which implies:

$$\sigma_{g_1}(a) = \sigma_{g_2}(a) \quad \forall a \in G.$$

Setting $a = e \in G$:

$$g_1 e = g_2 e \implies g_1 = g_2.$$

Thus, f is injective, which implies $\ker f = \{e\}$. By the First Isomorphism Theorem for groups:

$$G/\ker f \cong f(G) \implies G/\{e\} \cong f(G) \implies G \cong f(G).$$

Since $f(G) \leq \text{Sym}(G)$, G is isomorphic to the subgroup $f(G)$ of $\text{Sym}(G)$. □

3. Kernel of an Action, Stabilizers, and Orbits

Definition 3.1 (Kernel of an Action). Let $* : G \times A \rightarrow A$ be a group action. The **kernel** of the action $*$ is defined as:

$$\ker * = \{g \in G \mid g * a = a, \forall a \in A\}.$$

That is, $\ker *$ consists of all elements of G that fix every element of A .

Theorem 3.2. *For any group action $* : G \times A \rightarrow A$, the kernel is a normal subgroup of G , $\ker * \trianglelefteq G$.*

Proof. Since $e * a = a$ for all $a \in A$, $e \in \ker *$, so $\ker * \neq \emptyset$.

1) Closure under multiplication: Let $g_1, g_2 \in \ker *$. Then for any $a \in A$:

$$(g_1 g_2) * a = g_1 * (g_2 * a) = g_1 * a = a \implies g_1 g_2 \in \ker *.$$

2) Closure under inverses: Let $g \in \ker *$, so $g * a = a$ for all $a \in A$. Then:

$$g^{-1} * a = g^{-1} * (g * a) = (g^{-1} g) * a = e * a = a \implies g^{-1} \in \ker *.$$

Thus, $\ker * \leq G$. (Normality follows directly from Theorem 3.3 below since $\ker *$ is the kernel of a homomorphism.) □

Theorem 3.3. *If $f : G \rightarrow \text{Sym}(A)$ is the permutation representation corresponding to $*$, then:*

$$\ker f = \ker *$$

Proof. By definition of the kernel of a group homomorphism:

$$\begin{aligned}
 \ker f &= \{g \in G \mid f(g) = I_A\} \\
 &= \{g \in G \mid \sigma_g = I_A\} \\
 &= \{g \in G \mid \sigma_g(a) = I_A(a), \forall a \in A\} \\
 &= \{g \in G \mid g * a = a, \forall a \in A\} \\
 &= \ker *.
 \end{aligned}$$

□

Stabilizers and Orbits

Definition 3.4 (Stabilizer). Let G act on a set A and let $a \in A$. The **stabilizer** of a in G (also called the isotropy subgroup) is:

$$G_a = \{g \in G \mid g * a = a\}.$$

Theorem 3.5. For each $a \in A$, the stabilizer G_a is a subgroup of G , that is, $G_a \leq G$.

Proof. Since $e * a = a$, we have $e \in G_a$, so $G_a \neq \emptyset$.

1) Let $g_1, g_2 \in G_a$. Then $g_1 * a = a$ and $g_2 * a = a$. Consequently:

$$(g_1 g_2) * a = g_1 * (g_2 * a) = g_1 * a = a \implies g_1 g_2 \in G_a.$$

2) Let $g \in G_a$, so $g * a = a$. Acting by g^{-1} :

$$g^{-1} * a = g^{-1} * (g * a) = (g^{-1} g) * a = e * a = a \implies g^{-1} \in G_a.$$

By the two-step subgroup criterion, $G_a \leq G$.

□

Example 3.6 (Stabilizer under Conjugation is the Normalizer). If G acts on itself by conjugation ($g * a = gag^{-1}$ for $a \in G$), then for any $a \in G$:

$$\begin{aligned}
 G_a &= \{g \in G \mid g * a = a\} = \{g \in G \mid gag^{-1} = a\} \\
 &= \{g \in G \mid ga = ag\} = N(a),
 \end{aligned}$$

where $N(a)$ is the **normalizer** of a in G .

Definition 3.7 (Orbit). Let G act on A and let $a \in A$. The **orbit** of a under G is the set of all elements in A to which a can be moved by elements of G :

$$G \cdot a = \{x \in A \mid g * a = x \text{ for some } g \in G\} = \{g * a \mid g \in G\}.$$

Theorem 3.8 (Orbit-Stabilizer Theorem). Let G be a group acting on a set A , and let $a \in A$. There exists a bijection between the orbit $G \cdot a$ and the set G/G_a of all left cosets of G_a in G .

Proof. Let $G \cdot a = \{g * a \mid g \in G\}$ and $G/G_a = \{gG_a \mid g \in G\}$. Define the mapping:

$$\phi : G \cdot a \longrightarrow G/G_a, \quad \phi(g * a) = gG_a.$$

We establish that ϕ is well-defined, injective, and surjective:

- **Well-defined:** Suppose $g_1 * a = g_2 * a$. Then:

$$\begin{aligned} g_2^{-1} * (g_1 * a) &= g_2^{-1} * (g_2 * a) \implies (g_2^{-1}g_1) * a = e * a = a \\ &\implies g_2^{-1}g_1 \in G_a \\ &\implies g_1G_a = g_2G_a \\ &\implies \phi(g_1 * a) = \phi(g_2 * a). \end{aligned}$$

Thus, ϕ is independent of the choice of representative.

- **One-to-one (Injective):** Suppose $\phi(g_1 * a) = \phi(g_2 * a)$. Then:

$$\begin{aligned} g_1G_a = g_2G_a &\implies g_2^{-1}g_1 \in G_a \\ &\implies (g_2^{-1}g_1) * a = a \\ &\implies g_2 * ((g_2^{-1}g_1) * a) = g_2 * a \\ &\implies (g_2g_2^{-1}g_1) * a = g_2 * a \\ &\implies g_1 * a = g_2 * a. \end{aligned}$$

Thus ϕ is injective.

- **Onto (Surjective):** Let $t \in G/G_a$. Then $t = gG_a$ for some $g \in G$. For this g , the element $g * a \in G \cdot a$ satisfies $\phi(g * a) = gG_a = t$. Hence ϕ is surjective.

Therefore, ϕ is a bijection between $G \cdot a$ and G/G_a . □

Corollary 3.9. *If G is a finite group, then for any $a \in A$:*

$$|G \cdot a| = [G : G_a] = \frac{|G|}{|G_a|} \implies |G| = |G \cdot a| \cdot |G_a|$$

4. Conjugacy Classes and the Class Equation

Theorem 4.1 (Orbit Partitioning). *Let $* : G \times A \rightarrow A$ be a group action. Define a relation $\sim$ on A by:*

$$a \sim b \iff a = g * b \text{ for some } g \in G.$$

Then $\sim$ is an equivalence relation on A , and the equivalence classes are the orbits $G \cdot a$.

Proof. We check the three defining properties of an equivalence relation:

- 1) **Reflexive:** Since $e \in G$, $a = e * a$, so $a \sim a$ for all $a \in A$.
- 2) **Symmetric:** Let $a \sim b$. Then $a = g * b$ for some $g \in G$. Acting by g^{-1} :

$$g^{-1} * a = g^{-1} * (g * b) = (g^{-1}g) * b = e * b = b \implies b = g^{-1} * a \implies b \sim a.$$

- 3) **Transitive:** Let $a \sim b$ and $b \sim c$. Then there exist $g_1, g_2 \in G$ such that $a = g_1 * b$ and $b = g_2 * c$. Substituting gives:

$$a = g_1 * (g_2 * c) = (g_1g_2) * c.$$

Since $g_1g_2 \in G$, this implies $a \sim c$.

Thus, $\sim$ is an equivalence relation. The equivalence class containing $a \in A$ is:

$$\text{Cl}(a) = \{b \in A \mid a \sim b\} = \{g * a \mid g \in G\} = G \cdot a.$$

Since the equivalence classes partition A , we obtain:

$$A = \bigcup_{a \in A} \text{Cl}(a) = \bigcup_{a \in A} G \cdot a. \quad \square$$

Definition 4.2 (Conjugacy Class). Under the action of G on itself by conjugation ($g * a = gag^{-1}$), the orbit of $a \in G$ is called the **conjugacy class** of a , denoted $\text{Cl}(a)$:

$$\text{Cl}(a) = \{gag^{-1} \mid g \in G\}.$$

Exercise 4.3 (Conjugacy as an Equivalence Relation). Let G be a group. Two elements $a, b \in G$ are said to be conjugate (written $a \sim b$) if there exists $c \in G$ such that $b = cac^{-1}$. Prove directly that $\sim$ is an equivalence relation on G .

Solution. We verify the three equivalence relation axioms:

- 1) **Reflexivity:** For any $a \in G$, $a = eae^{-1}$ (since $e \in G$), so $a \sim a$.
- 2) **Symmetry:** Let $a \sim b$. Then there exists $c \in G$ such that $a = bcb^{-1}$. Multiplying on the left by c^{-1} and on the right by c :

$$c^{-1}ac = c^{-1}(bcb^{-1})c = (c^{-1}c)b(c^{-1}c) = ebe = b.$$

Writing $c = (c^{-1})^{-1}$, this gives $b = c^{-1}a(c^{-1})^{-1}$. Since $c^{-1} \in G$, we have $b \sim a$.

- 3) **Transitivity:** Let $a \sim b$ and $b \sim c$. Then there exist $d_1, d_2 \in G$ such that:

$$a = d_1bd_1^{-1} \quad \text{and} \quad b = d_2cd_2^{-1}.$$

Substituting b into the expression for a :

$$a = d_1(d_2cd_2^{-1})d_1^{-1} = (d_1d_2)c(d_2^{-1}d_1^{-1}) = (d_1d_2)c(d_1d_2)^{-1}.$$

Since $d_1d_2 \in G$, it follows that $a \sim c$.

Thus, conjugacy is an equivalence relation on G . $\square$

Proposition 4.4 (Conjugacy Classes and the Center). Let G be a group and let $a \in G$. Then:

- 1) $\text{Cl}(a) = \{a\} \iff a \in Z(G)$.
- 2) $N(a) = G \iff a \in Z(G)$.

Proof. Part 1: ($\implies$) Let $\text{Cl}(a) = \{a\}$. For any $g \in G$, $gag^{-1} \in \text{Cl}(a) = \{a\}$, so $gag^{-1} = a$. Multiplying on the right by g yields $ga = ag$. Since g was arbitrary, a commutes with every element of G , so $a \in Z(G)$.

($\impliedby$) Conversely, suppose $a \in Z(G)$. Then for all $g \in G$, $ga = ag$, so $gag^{-1} = a$. Thus:

$$\text{Cl}(a) = \{gag^{-1} \mid g \in G\} = \{agg^{-1} \mid g \in G\} = \{a\}.$$

Part 2: ($\implies$) Suppose $N(a) = G$. Then every $g \in G$ satisfies $g \in N(a)$, meaning $gag^{-1} = a$, or equivalently $ga = ag$. Thus $a \in Z(G)$.

($\impliedby$) Conversely, if $a \in Z(G)$, then $ga = ag$ for all $g \in G$, which means $g \in N(a)$ for all $g \in G$. Hence $N(a) = G$. $\square$

The Class Equation

For any finite group G , because the distinct conjugacy classes partition G , we have:

$$|G| = \sum_{a \in G} |\text{Cl}(a)|.$$

By the Orbit-Stabilizer Theorem, $|\text{Cl}(a)| = [G : N(a)] = \frac{|G|}{|N(a)|}$. Separating the singleton conjugacy classes (which correspond precisely to elements of the center $Z(G)$) from the non-central classes yields:

The Class Equation

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right| = |Z(G)| + \sum_{a \notin Z(G)} [G : N(a)]$$

where the sum runs over one representative a from each conjugacy class not contained in $Z(G)$.

Applications of the Class Equation to p -Groups

Theorem 4.5 (Center of a p -Group is Non-Trivial). *Let G be a finite group of order $|G| = p^n$, where p is a prime and $n \geq 1$. Then $|Z(G)| \geq p$, that is, the center of a non-trivial p -group is non-trivial.*

Proof. Let $|Z(G)| = r$. By the class equation:

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} [G : N(a)]. \quad (\text{A})$$

For any $a \notin Z(G)$, the normalizer $N(a)$ is a proper subgroup of G , so $N(a) \subsetneq G$. By Lagrange's Theorem, $|N(a)|$ divides $|G| = p^n$, so $|N(a)| = p^{\alpha_a}$ with $\alpha_a < n$. Therefore:

$$[G : N(a)] = \frac{|G|}{|N(a)|} = \frac{p^n}{p^{\alpha_a}} = p^{n-\alpha_a}, \quad \text{where } n - \alpha_a \geq 1.$$

Thus, $p \mid [G : N(a)]$ for each term in the summation, which implies:

$$p \mid \sum_{a \notin Z(G)} [G : N(a)].$$

Since $|G| = p^n$, we also have $p \mid |G|$. Rewriting the class equation as:

$$|Z(G)| = |G| - \sum_{a \notin Z(G)} [G : N(a)],$$

it follows that $p \mid |Z(G)|$. Since the identity element $e \in Z(G)$, the center is non-empty, so $|Z(G)| \geq 1$. Because $|Z(G)|$ is a positive integer divisible by the prime p , we conclude that:

$$|Z(G)| \geq p. \quad \square$$

Theorem 4.6 (Groups of Order p^2). *Every group G of order p^2 , where p is a prime, is abelian.*

Proof. By Theorem 4.5, $|Z(G)| \geq p$, and by Lagrange's Theorem, $|Z(G)|$ must divide $|G| = p^2$. Hence:

$$|Z(G)| = p \quad \text{or} \quad |Z(G)| = p^2.$$

Suppose for contradiction that $|Z(G)| = p$. Then the quotient group has order:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{p^2}{p} = p.$$

Since any group of prime order is cyclic, $G/Z(G)$ is cyclic. It is a standard result that if $G/Z(G)$ is cyclic, then G is abelian. But if G is abelian, then $Z(G) = G$, so $|Z(G)| = p^2$, which contradicts $|Z(G)| = p$.

Hence, $|Z(G)| \neq p$. The only remaining possibility is $|Z(G)| = p^2 = |G|$, which implies $Z(G) = G$. Therefore, G is abelian. $\square$

Remark 4.7. If H is a subgroup of G such that $H \leq Z(G)$, then H is normal in G , $H \trianglelefteq G$. This holds because every element of H commutes with all elements of G , so $ghg^{-1} = h \in H$ for all $g \in G, h \in H$.

Conjugacy Classes in Symmetric Groups S_n

Two permutations in S_n are conjugate if and only if they have the same cycle decomposition structure (cycle type).

Theorem 4.8 (Size of a Conjugacy Class in S_n). Let $\sigma \in S_n$ have cycle type consisting of k_i cycles of length m_i for $i = 1, 2, \dots, r$ (including 1-cycles), such that $\sum_{i=1}^r k_i m_i = n$. Then the number of conjugates of σ in S_n is:

$$|\text{Cl}(\sigma)| = \frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \cdots (k_r! m_r^{k_r})}$$

Example 4.9 (Calculations in S_4). Here $n = 4$, so $4! = 24$.

- **Transpositions (cycles of length 2):** The cycle type is $2^1 1^2$ ($k_1 = 1, m_1 = 2$ and $k_2 = 2, m_2 = 1$).

$$|\text{Cl}((12))| = \frac{4!}{(1! \cdot 2^1)(2! \cdot 1^2)} = \frac{24}{2 \times 2} = 6.$$

Explicitly: $\text{Cl}((12)) = \{(12), (13), (14), (23), (24), (34)\}$.

- **3-cycles:** The cycle type is $3^1 1^1$ ($k_1 = 1, m_1 = 3$ and $k_2 = 1, m_2 = 1$).

$$|\text{Cl}((123))| = \frac{4!}{(1! \cdot 3^1)(1! \cdot 1^1)} = \frac{24}{3 \times 1} = 8.$$

Example 4.10 (Calculations in S_5). Here $n = 5$, so $5! = 120$.

- **Transpositions (cycles of length 2):** The cycle type is $2^1 1^3$ ($k_1 = 1, m_1 = 2$ and $k_2 = 3, m_2 = 1$).

$$|\text{Cl}((12))| = \frac{5!}{(1! \cdot 2^1)(3! \cdot 1^3)} = \frac{120}{2 \times 6} = \frac{120}{12} = 10.$$

5. Introduction to Ring Theory

Definition 5.1 (Ring). A mathematical system $(R, +, \cdot)$ consisting of a non-empty set R equipped with two binary operations '+' (addition) and ' $\cdot$ ' (multiplication) is called a **ring** if it satisfies:

1) $(R, +)$ is an abelian group:

- Closure: $a + b \in R$ for all $a, b \in R$.
- Associativity: $(a + b) + c = a + (b + c)$ for all $a, b, c \in R$.
- Additive identity: There exists an element $0 \in R$ such that $a + 0 = 0 + a = a$ for all $a \in R$.
- Additive inverse: For each $a \in R$, there exists $-a \in R$ such that $a + (-a) = (-a) + a = 0$.
- Commutativity: $a + b = b + a$ for all $a, b \in R$.

2) $(R, \cdot)$ is a semigroup:

- Closure: $a \cdot b \in R$ for all $a, b \in R$.
- Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$.

3) Distributivity of multiplication over addition: For all $a, b, c \in R$:

$$\begin{aligned} a \cdot (b + c) &= a \cdot b + a \cdot c && \text{(Left distributive law),} \\ (a + b) \cdot c &= a \cdot c + b \cdot c && \text{(Right distributive law).} \end{aligned}$$

Definition 5.2 (Special Classes of Rings).

- 1) Zero Element:** The identity of $(R, +)$ is called the **zero element** of the ring, denoted 0 .
- 2) Ring with Unity:** If $(R, \cdot)$ has an identity element $1 \in R$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in R$, then R is called a **ring with unity** (or identity).
- 3) Commutative Ring:** If multiplication is commutative ($a \cdot b = b \cdot a$ for all $a, b \in R$), then R is called a **commutative ring**.

Twelve Standard Examples of Rings

- 1) Number Systems:** The systems $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, and $(\mathbb{C}, +, \cdot)$ under standard addition and multiplication are all commutative rings with unity 1.
- 2) Integers Modulo n :** For any integer $n \geq 2$, $(\mathbb{Z}_n, +_n, \times_n)$ is a commutative ring with unity $[1]$.
- 3) Quadratic Ring of Integers:** For a square-free integer p , $\mathbb{Z}[\sqrt{p}] = \{a + b\sqrt{p} \mid a, b \in \mathbb{Z}\}$ forms a commutative ring with unity 1.
- 4) Ring of Gaussian Integers:** The set $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ under complex addition and multiplication is a commutative ring with unity 1.
- 5) Real Matrix Ring:** For $n \geq 2$, $M_n(\mathbb{R})$ (the set of all $n \times n$ real matrices) under matrix addition and matrix multiplication satisfies:
 - $(M_n(\mathbb{R}), +)$ is an abelian group with zero matrix $0_{n \times n}$.
 - $(M_n(\mathbb{R}), \cdot)$ is a semigroup with identity I_n .

- Matrix multiplication distributes over matrix addition: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.

Since matrix multiplication is generally non-commutative, the ring $(M_n(\mathbb{R}), +, \cdot)$ is a **non-commutative ring with unity**.

6) **Integer Matrix Ring:** $M_n(\mathbb{Z})$ is similarly a non-commutative ring with unity I_n .

7) **A Constant 2×2 Matrix Ring with Non-Standard Unity:** Consider the set:

$$A = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix} \mid x \in \mathbb{R} \right\}.$$

Under standard matrix addition, $(A, +)$ is an abelian group. Under matrix multiplication:

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} y & y \\ y & y \end{bmatrix} = \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} = \begin{bmatrix} y & y \\ y & y \end{bmatrix} \begin{bmatrix} x & x \\ x & x \end{bmatrix}.$$

Thus, multiplication is commutative on A . To find the unity $E = \begin{bmatrix} e & e \\ e & e \end{bmatrix} \in A$:

$$\begin{bmatrix} x & x \\ x & x \end{bmatrix} \begin{bmatrix} e & e \\ e & e \end{bmatrix} = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \implies \begin{bmatrix} 2xe & 2xe \\ 2xe & 2xe \end{bmatrix} = \begin{bmatrix} x & x \\ x & x \end{bmatrix} \implies 2xe = x \implies e = \frac{1}{2}.$$

Hence, the unity of the ring A is:

$$E = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}.$$

Furthermore, for any non-zero element $X = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$ ($x \neq 0$), its multiplicative inverse $Y = \begin{bmatrix} y & y \\ y & y \end{bmatrix}$ satisfies:

$$XY = E \implies 2xy = \frac{1}{2} \implies y = \frac{1}{4x} \implies Y = \begin{bmatrix} \frac{1}{4x} & \frac{1}{4x} \\ \frac{1}{4x} & \frac{1}{4x} \end{bmatrix}.$$

Thus $(A \setminus \{0\}, \cdot)$ is an abelian group, making A a field!

8) **Matrix Ring with Projection Unity:** The set:

$$A = \left\{ \begin{bmatrix} x & 0 \\ 0 & 0 \end{bmatrix} \mid x \in \mathbb{R} \right\}$$

is a commutative ring with unity $E = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

9) **The Trivial Ring:** The singleton set $R = \{0\}$ with $0 + 0 = 0$ and $0 \cdot 0 = 0$ is a ring with unity 0 , called the **trivial ring** (or **zero ring**). Any ring with $R \neq \{0\}$ is called a **non-trivial ring**.

10) **Polynomial Ring over $\mathbb{Z}$:** The set $\mathbb{Z}[x]$ of all polynomials with integer coefficients is a commutative ring with unity 1 .

11) **Polynomial Rings over Fields:** The sets $\mathbb{Q}[x]$, $\mathbb{R}[x]$, and $\mathbb{C}[x]$ are all commutative rings with unity 1 .

12) Ring of Continuous Functions: Let $C[a, b]$ denote the set of all continuous real-valued functions defined on the closed interval $[a, b]$. Under pointwise addition and multiplication:

$$(f + g)(x) = f(x) + g(x), \quad (f \cdot g)(x) = f(x) \cdot g(x) \quad \forall x \in [a, b],$$

$C[a, b]$ is a commutative ring with unity $e(x) \equiv 1$ for all $x \in [a, b]$.

Fundamental Algebraic Properties of a Ring

Theorem 5.3 (Ring Identities). *Let $(R, +, \cdot)$ be a ring. Then for all $a, b, c \in R$:*

- 1) $a \cdot 0 = 0 \cdot a = 0$
- 2) $a(-b) = (-a)b = -(ab)$
- 3) $(-a)(-b) = ab$
- 4) $a(b - c) = ab - ac$
- 5) $(b - c)a = ba - ca$

Proof. **Part 1:** In $(R, +)$, $0 + 0 = 0$. By left distributivity:

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding the additive identity 0 to the left-hand side gives $0 + a \cdot 0 = a \cdot 0 + a \cdot 0$. By the left cancellation law in the abelian group $(R, +)$:

$$a \cdot 0 = 0.$$

Similarly, $(0 + 0) \cdot a = 0 \cdot a + 0 \cdot a \implies 0 \cdot a = 0$.

Part 2: By distributivity and Part 1:

$$a(-b) + ab = a(-b + b) = a \cdot 0 = 0.$$

Since the additive inverse of ab in $(R, +)$ is unique and denoted $-(ab)$, it follows that $a(-b) = -(ab)$. Similarly:

$$(-a)b + ab = (-a + a)b = 0 \cdot b = 0 \implies (-a)b = -(ab).$$

Part 3: Applying Part 2 twice:

$$(-a)(-b) = -((-a)b) = -(-(ab)) = ab.$$

Part 4: Recalling that $b - c = b + (-c)$ and using distributivity and Part 2:

$$a(b - c) = a(b + (-c)) = ab + a(-c) = ab + (-(ac)) = ab - ac.$$

Part 5: Similarly, by right distributivity:

$$(b - c)a = (b + (-c))a = ba + (-c)a = ba + (-(ca)) = ba - ca. \quad \square$$

6. Zero Divisors and Integral Domains

Definition 6.1 (Zero Divisors). Let R be a ring. A non-zero element $a \in R$ is called a **zero divisor** (or divisor of zero) if there exists a non-zero element $b \in R$ such that:

$$ab = 0 \quad \text{or} \quad ba = 0.$$

More specifically:

- $a \neq 0$ is a **left zero divisor** if there exists $b \neq 0$ such that $ab = 0$.
- $b \neq 0$ is a **right zero divisor** if there exists $a \neq 0$ such that $ab = 0$.

Example 6.2 (Zero Divisors in $\mathbb{Z}_6$). In the ring $(\mathbb{Z}_6, +_6, \times_6)$, the non-zero elements $[2], [3], [4]$ are zero divisors because:

$$[2] \times_6 [3] = [0], \quad [3] \times_6 [4] = [0].$$

In contrast, in the rings $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$, there are no zero divisors.

Definition 6.3 (Integral Domain). A commutative ring R with unity ($1 \neq 0$) is called an **integral domain** if it has no zero divisors, that is, for all $a, b \in R$:

$$ab = 0 \implies a = 0 \quad \text{or} \quad b = 0.$$

Examples and Non-Examples of Integral Domains.

1) Examples:

- The number systems $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, and $(\mathbb{C}, +, \cdot)$ are integral domains.
- $(\mathbb{Z}_p, +_p, \times_p)$ is an integral domain if and only if p is a prime.
- The polynomial rings $\mathbb{Z}[x], \mathbb{Q}[x], \mathbb{R}[x]$, and $\mathbb{C}[x]$ are integral domains.
- The Gaussian integers $\mathbb{Z}[i]$ form an integral domain.

2) Non-Examples:

- For composite n , $(\mathbb{Z}_n, +_n, \times_n)$ is not an integral domain (for example, in $\mathbb{Z}_4$, $[2] \times_4 [2] = [0]$ with $[2] \neq [0]$).
- For $n \geq 2$, $M_n(\mathbb{R})$ is not an integral domain because it is non-commutative and possesses zero divisors (e.g., $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^2 = 0$).

Theorem 6.4 (Cancellation Law in Integral Domains). A commutative ring R is an integral domain if and only if the cancellation law holds under multiplication, that is, for all $a, b, c \in R$ with $a \neq 0$:

$$ab = ac \implies b = c.$$

Proof. ($\implies$) Let R be an integral domain, and let $ab = ac$ with $a \neq 0$. Then:

$$ab - ac = 0 \implies a(b - c) = 0.$$

Since R is an integral domain and $a \neq 0$, R contains no zero divisors, which forces $b - c = 0$, so $b = c$.

($\impliedby$) Conversely, assume that the cancellation law holds in R . Suppose $ab = 0$ with $a \neq 0$. By Theorem 5.3, $a \cdot 0 = 0$, so:

$$ab = a \cdot 0.$$

Since $a \neq 0$, the cancellation law immediately gives $b = 0$. Therefore, R has no zero divisors, so R is an integral domain. $\square$

7. Fields, Division Rings, and Ring Characteristic

Definition 7.1 (Field). A commutative ring $(F, +, \cdot)$ with unity ($1 \neq 0$) is called a **field** if every non-zero element has a multiplicative inverse in F , that is:

$$\forall a \in F \setminus \{0\}, \quad \exists a^{-1} \in F \quad \text{such that} \quad a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

Equivalently, $(F, +, \cdot)$ is a field if:

- 1) $(F, +)$ is an abelian group.
- 2) $(F \setminus \{0\}, \cdot)$ is an abelian group.
- 3) Distributive laws hold.

Definition 7.2 (Division Ring / Skew-Field). A ring $(R, +, \cdot)$ with unity ($1 \neq 0$) is called a **division ring** (or **skew-field**) if every non-zero element has a multiplicative inverse. (Multiplication need not be commutative.)

Examples and Non-Examples of Fields.

- $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, and $(\mathbb{C}, +, \cdot)$ are fields.
- $(\mathbb{Z}_p, +_p, \times_p)$ is a field if and only if p is prime.
- $(\mathbb{Z}, +, \cdot)$ is **not** a field because only ± 1 have multiplicative inverses in $\mathbb{Z}$.
- $(\mathbb{Z}_n, +_n, \times_n)$ for composite n is not a field.

Theorem 7.3. *Every field is an integral domain.*

Proof. Let F be a field. Then F is a commutative ring with unity $1 \neq 0$. To prove that F is an integral domain, we must show that it has no zero divisors.

Let $a, b \in F$ such that $ab = 0$. If $a = 0$, we are done. If $a \neq 0$, then because F is a field, the multiplicative inverse $a^{-1} \in F$ exists. Multiplying both sides by a^{-1} :

$$\begin{aligned} a^{-1}(ab) &= a^{-1} \cdot 0 \\ (a^{-1}a)b &= 0 \\ 1 \cdot b &= 0 \\ b &= 0. \end{aligned}$$

Thus, $ab = 0$ implies $a = 0$ or $b = 0$. Therefore, F has no zero divisors, so F is an integral domain. $\square$

Remark 7.4. The converse of Theorem 7.3 does not hold in general: $(\mathbb{Z}, +, \cdot)$ is an integral domain, but it is not a field. However, in the finite setting, the converse is universally true!

Theorem 7.5 (Finite Integral Domains are Fields). *Every finite integral domain is a field.*

Proof. Let D be a finite integral domain. Since D is finite, its elements can be listed as:

$$D = \{x_1, x_2, \dots, x_n\},$$

where $x_1, \dots, x_n$ are distinct elements. Let $a \in D$ be any non-zero element ($a \neq 0$). Consider the set of products:

$$aD = \{ax_1, ax_2, \dots, ax_n\}.$$

Since D is closed under multiplication, $aD \subseteq D$. We claim that all n elements of aD are distinct:

$$ax_i = ax_j \implies ax_i - ax_j = 0 \implies a(x_i - x_j) = 0.$$

Because D is an integral domain and $a \neq 0$, D contains no zero divisors, which forces $x_i - x_j = 0$, so $x_i = x_j$.

Hence, if $i \neq j$, then $ax_i \neq ax_j$. Therefore, aD contains n distinct elements of D . Since D has exactly n elements, by the Pigeonhole Principle:

$$aD = D.$$

Since D is an integral domain with unity 1, we have $1 \in D = aD$. Thus, there exists some $x_k \in D$ such that:

$$ax_k = 1.$$

Because D is commutative, $ax_k = x_k a = 1$, which means $x_k = a^{-1}$. Thus, every non-zero element of D possesses a multiplicative inverse in D . Hence, D is a field. $\square$

Corollary 7.6. $(\mathbb{Z}_p, +_p, \times_p)$ is a field if and only if p is a prime number.

Characteristic of a Ring

Definition 7.7 (Characteristic). Let $(R, +, \cdot)$ be a ring. If there exists a least positive integer n such that:

$$n \cdot a = \underbrace{a + a + \dots + a}_{n \text{ times}} = 0 \quad \forall a \in R,$$

then n is called the **characteristic** of the ring R , denoted $\text{char}(R) = n$. If no such positive integer exists, R is said to have **characteristic zero** (or infinite characteristic), denoted $\text{char}(R) = 0$.

Examples of Characteristics.

- $\text{char}(\mathbb{Z}) = 0$, $\text{char}(\mathbb{Q}) = 0$, $\text{char}(\mathbb{R}) = 0$, $\text{char}(\mathbb{C}) = 0$.
- $\text{char}(\mathbb{Z}_n) = n$ for any $n \geq 2$.
- $\text{char}(\mathbb{Z}_p) = p$ for any prime p .

Theorem 7.8 (Characteristic Determined by Unity). Let R be a ring with unity 1.

- 1) If the order of 1 under addition in $(R, +)$ is finite, say $O(1) = n$, then $\text{char}(R) = n$.
- 2) If the order of 1 under addition is infinite, then $\text{char}(R) = 0$.

Proof. Part 1: Let $O(1) = n$ in $(R, +)$. Then n is the least positive integer such that $n \cdot 1 = 0$. For any arbitrary element $a \in R$:

$$n \cdot a = \underbrace{a + a + \dots + a}_{n \text{ times}} = \underbrace{(1 + 1 + \dots + 1)}_{n \text{ times}} a = (n \cdot 1)a = 0 \cdot a = 0.$$

Thus, $n \cdot a = 0$ for all $a \in R$. Furthermore, since n is the least positive integer satisfying $n \cdot 1 = 0$, it is the least positive integer annihilating all elements of R . Therefore, $\text{char}(R) = n$.

Part 2: If $O(1)$ is infinite, then there exists no positive integer n such that $n \cdot 1 = 0$. Consequently, no positive integer can annihilate all elements of R , so $\text{char}(R) = 0$. $\square$

Theorem 7.9 (Characteristic of an Integral Domain). *The characteristic of an integral domain (and hence of any field) is either 0 or a prime number.*

Proof. Let D be an integral domain. If $\text{char}(D) = 0$, the theorem holds.

Now suppose $\text{char}(D) = n > 0$. We must show that n is a prime number. Suppose for contradiction that n is composite. Then there exist integers n_1, n_2 such that:

$$n = n_1 n_2, \quad \text{where } 1 < n_1, n_2 < n.$$

Since $\text{char}(D) = n$, by Theorem 7.8 we have $n \cdot 1 = 0$. Therefore:

$$(n_1 n_2) \cdot 1 = 0 \implies (n_1 \cdot 1)(n_2 \cdot 1) = 0.$$

Because D is an integral domain, it has no zero divisors. Thus, the above product can only be zero if:

$$n_1 \cdot 1 = 0 \quad \text{or} \quad n_2 \cdot 1 = 0.$$

However, this contradicts the definition of n as the *least* positive integer such that $n \cdot 1 = 0$, since both $n_1 < n$ and $n_2 < n$.

Hence, n cannot be composite. Therefore, n must be a prime number. $\square$